

The Roll and Calendar Spreads

Managing The Expiration Cycle

FINM 37000 - Futures And Related Derivatives

Eric Patterson, Summer 2026

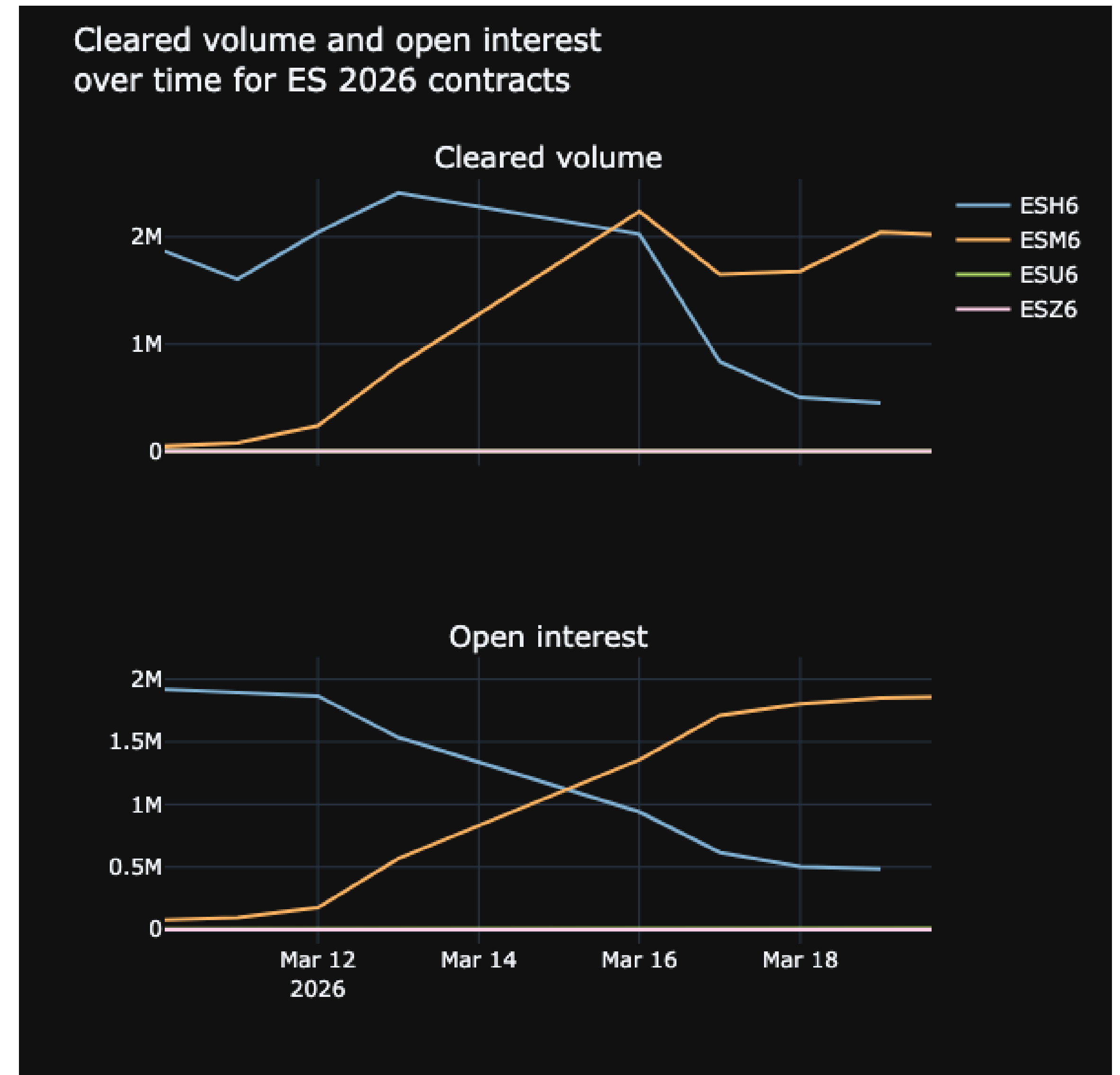
Expiration Cycle

Month	Code	Soybeans	Corn	Crude	Ten Years	E-mini S&P	3 Month SOFR
May	K	ZSK6	ZCK6	April Settle			SRK6
June	M			CLM6	ZNM6	ESM6	SR3M6
July	N	ZSN6	ZCN6	CLN6			SRN6
August	Q	ZSQ6		CLQ6			SRQ6
September	U	ZSU6	ZCU6	CLU6	ZNU6	ESU6	SR3U6
October	V			CLV6			SRV6
November	X	ZSX6		CLX6			
December	Z		ZCZ6	CLZ6	ZNZ6	ESZ6	SR3Z6
January	F	ZSF7		CLF7			
February	G			CLG7			
March	H	ZSH7	ZCH7	CLH7	(ZNH7)	ESH7	SR3H7
April	J			CLJ7			
Total		8/15	8/9	10 yrs+2mo		21 quarters	39 qtrs/6 serial

Rolling Positions

Shifting Expiration Exposure

- Suppose you are short 1000 E-mini S&P June futures (ESH6) as a hedge for a portfolio of stocks.
- Before those futures expire on March 20th, roll them to June (ESM6).
- Buy ESH6, Sell ESM6.

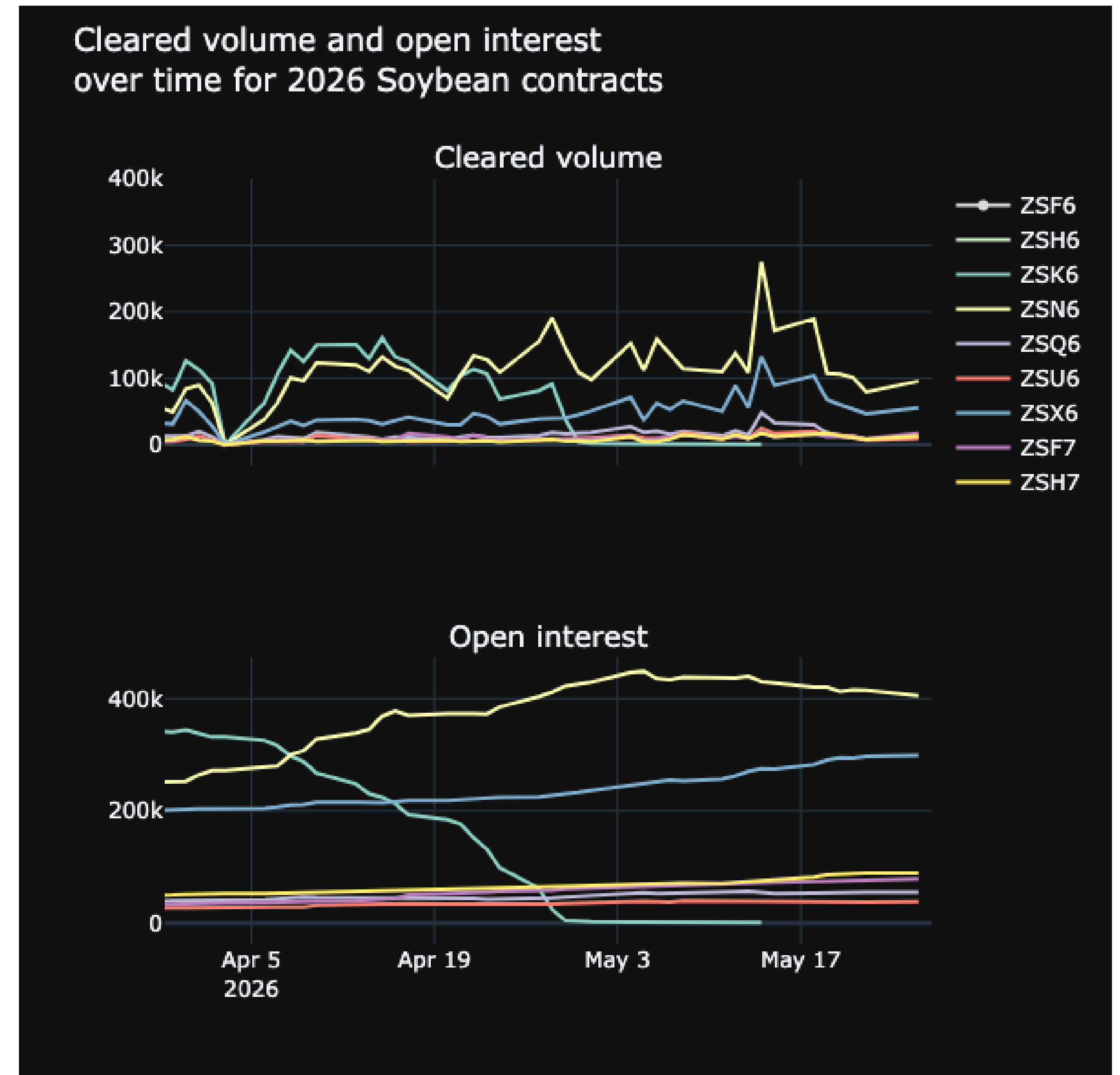


Caption

Rolling Positions

Shifting Expiration Exposure

- Suppose you are short 50 July Soybean futures (ZSN6) as a part of a strategy to benefit from falling Soybean prices.
- Before those futures expire on July 14th, roll them to another month.
 - Which month would you roll to?



Caption

Rolling Positions

Leg Execution

- Suppose hedger wants to sell 50 November soybeans
- Not enough liquidity at the top of the book
- Possibility 1:
 - Hit ZSX6 Bid 1040 for 10
 - Hit next 1039'6 for 20
 - Hit 1039'4 for 20
 - Sold 50 but missed best bid 1040
 - $20 \times 0.25 \times 5000 = \250
 - $20 \times 0.5 \times 5000 = \500

ZSX6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		1040'2	10
10	1040'0		
20	1039'6		
20	1039'4		

Rolling Positions

Leg Executions

- Roll short 1000 S&P E-Mini June futures (ESH6)
- Possibility 1:
 - Lift offer 5500.75 ESH6
 - Closes out ESH6 position
 - Hit bid 5540.75 ESM6

ESH6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		5500.75	8000

ESM6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
2000	5540.75		

Calendar Spreads

Soybean Example

- ZSN6-ZSX6
 - Leg 1: +1 ZSN6
 - Leg 2: -1 ZSX6
 - Example: Trade 1 spread for 10'0
 - Typical leg price assignment: most recent price from a leg, say leg 1
 - ZSN6 assigned last traded 1050'0
 - ZSX6 assigned ZSN6-Spread = 1040'0

Calendar spreads

How trading exchange-listed spreads improves execution

- Suppose hedger wants to sell 50 November soybeans, but it's less liquid than July

ZSN6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	1050'0		

- Possibility 2:

- Lift 50 spreads offered 10
- Hit 50 July bids 1050'0
- Left short 50 ZSX6 at 1040
- Can you execute both trades?

ZSN6-ZSX6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		10'0	50

ZSX6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
10	1040'0		

Calendar spreads

How trading exchange-listed spreads improves execution

- Suppose hedger wants to sell 50 November soybeans, but it's less liquid than July

ZSN6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
100	1050'0		

- Possibility 3:

- Exchange implies 50 in ZSX6

ZSN6-ZSX6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
		10'0	50

- Hit 1040'0 bid in ZSX6

ZSX6

Bid Quantity	Bid Price	Ask Price	Ask Quantity
10 + 50(i)	1040'0		

- First 10 filled by real orders
 - Next 40 implied

Calendar spreads

Benefits

- Improve leg execution
 - Often reduced tick size
- Reduced margin requirements (offsetting risk)
- Simplified roll

Other baskets

Reducing execution costs for multi-leg positions

- Packs
 - 4 consecutive quarterly futures
 - Price is relative to each futures' previous settle
- Bundles
 - At least 4 consecutive quarterly futures
 - Price as average of the futures
- Strips
 - At least 2 legs of same product with different expirations
 - Various pricing styles